



Vergleich von linear mixed models und mixed effects ANOVA zur Analyse des klassischen Therapie-Evaluationsdesigns

Comparison of linear mixed models and mixed effects ANOVA for the analysis of the classical therapy evaluation design

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Where We Are

Week	Lecture title
1	Introduction: reproducibility and simulation
2	Everything is Regression
3	Means and Intercepts
4	Slopes and Change
5	t-tests and one-way ANOVA
6	Mixed ANOVA (and multilevel models)
7	ANCOVA and (good and bad) controls
8	Linear and logistic regression
9	Latent variable modelling
10	Power analysis and the null hypothesis
11	Measurement error
12	Missingness
13	Everything is Still Regression

Reading

Chapter 11 ('Everything is Regression') of
the Monte Carlo Simulation textbook
(Hussey & Cummins, 2026)

simulations.tidyverse.se

In R and tidyverse



Understanding Statistics

*through Monte Carlo
simulations*

Ian Hussey & Jamie Cummins

Learning Objectives

Describe the components of the classical therapy evaluation design: AKA a randomized controlled trial

Distinguish between fixed effects and random effects, and between random intercepts and random slopes

Contrast how mixed ANOVA vs. multilevel models handle fixed vs. random effects

Demonstrate how failing to model important random effects can inflate Type I error

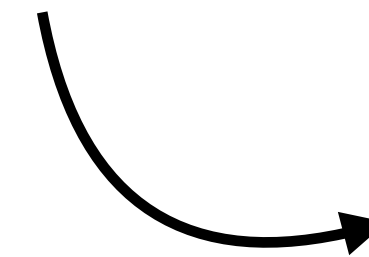
Examples Use Simulated Data

All code, data, and plots are on the class GitHub page.

English and German slides on GitHub too.

We use simulations and simulated data.

Class GitHub



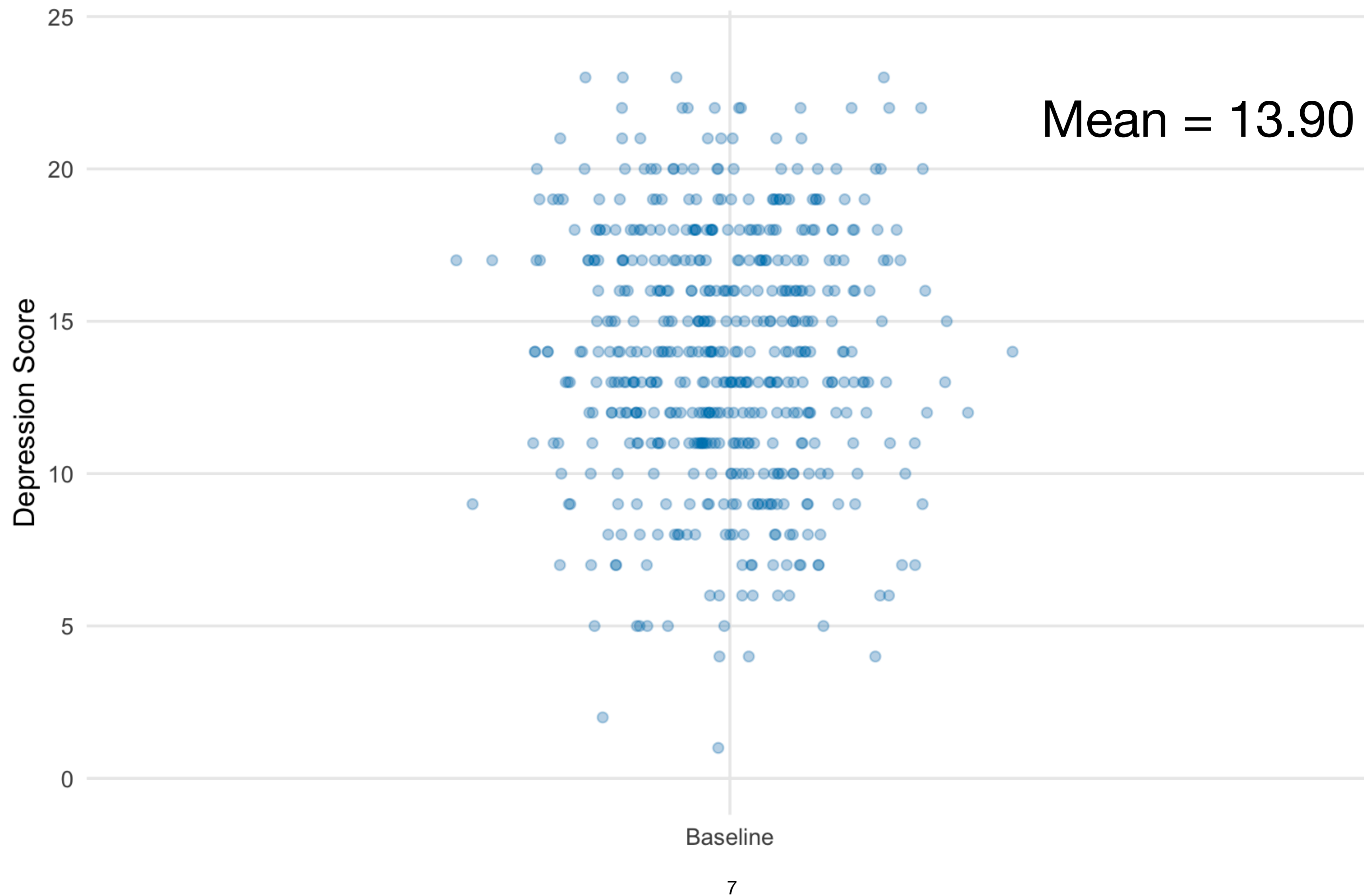
Reminder

Statistical Notation

dependent variable ~ independent variable(s)

Reminder

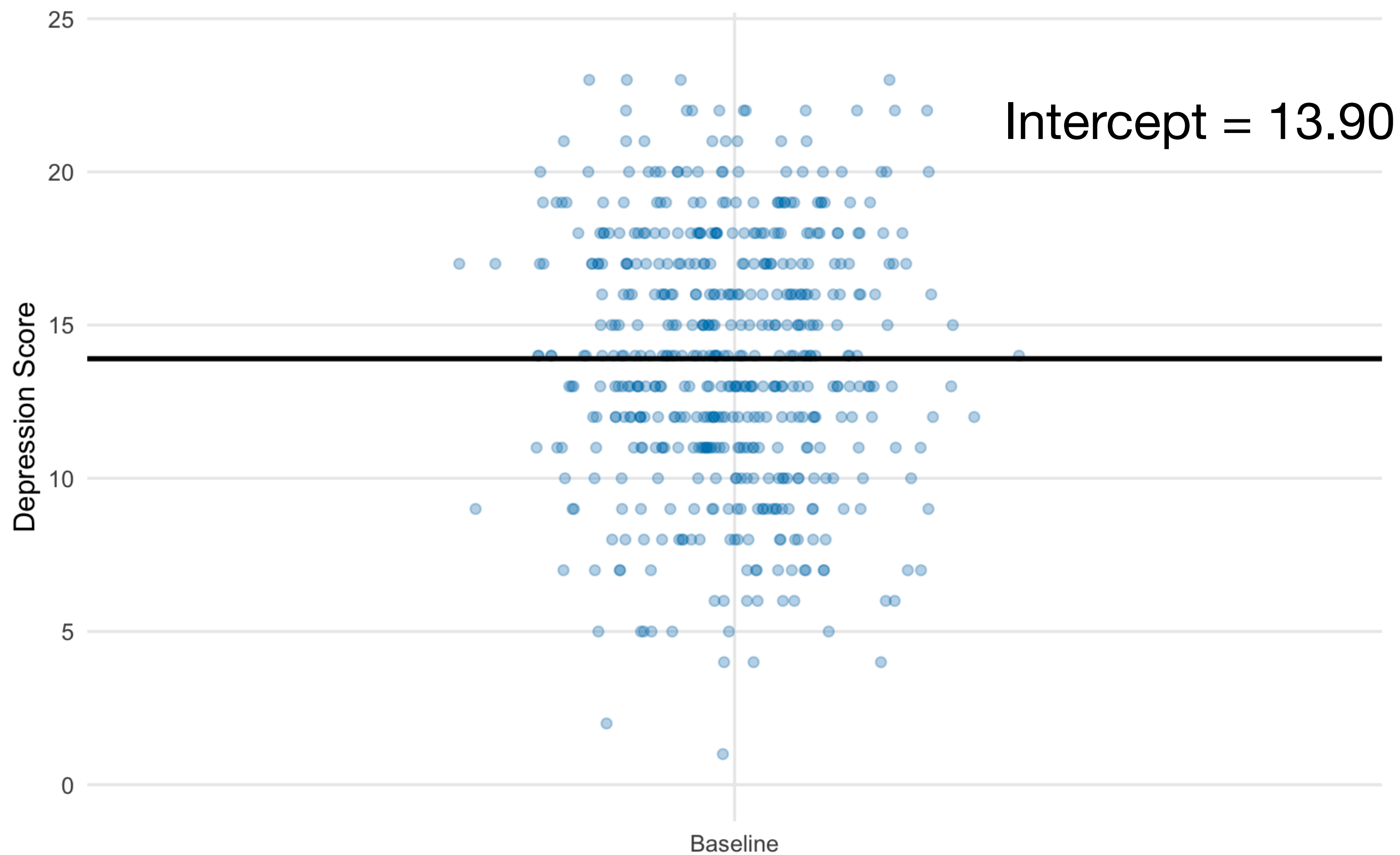
A Mean Is A (Intercept-Only) Model



Reminder

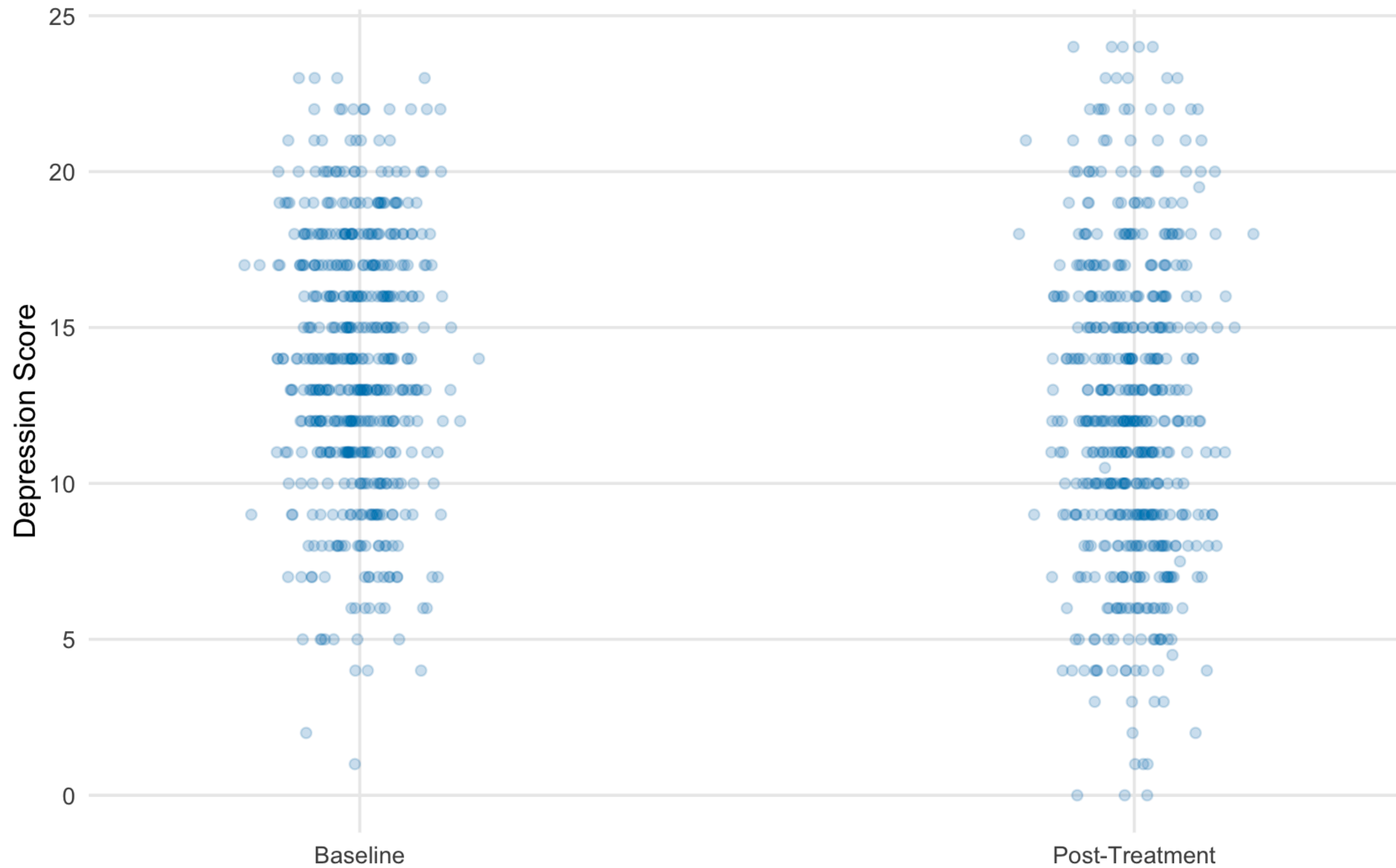
depression ~ 1

A Mean Is A (Intercept-Only) Model



Reminder

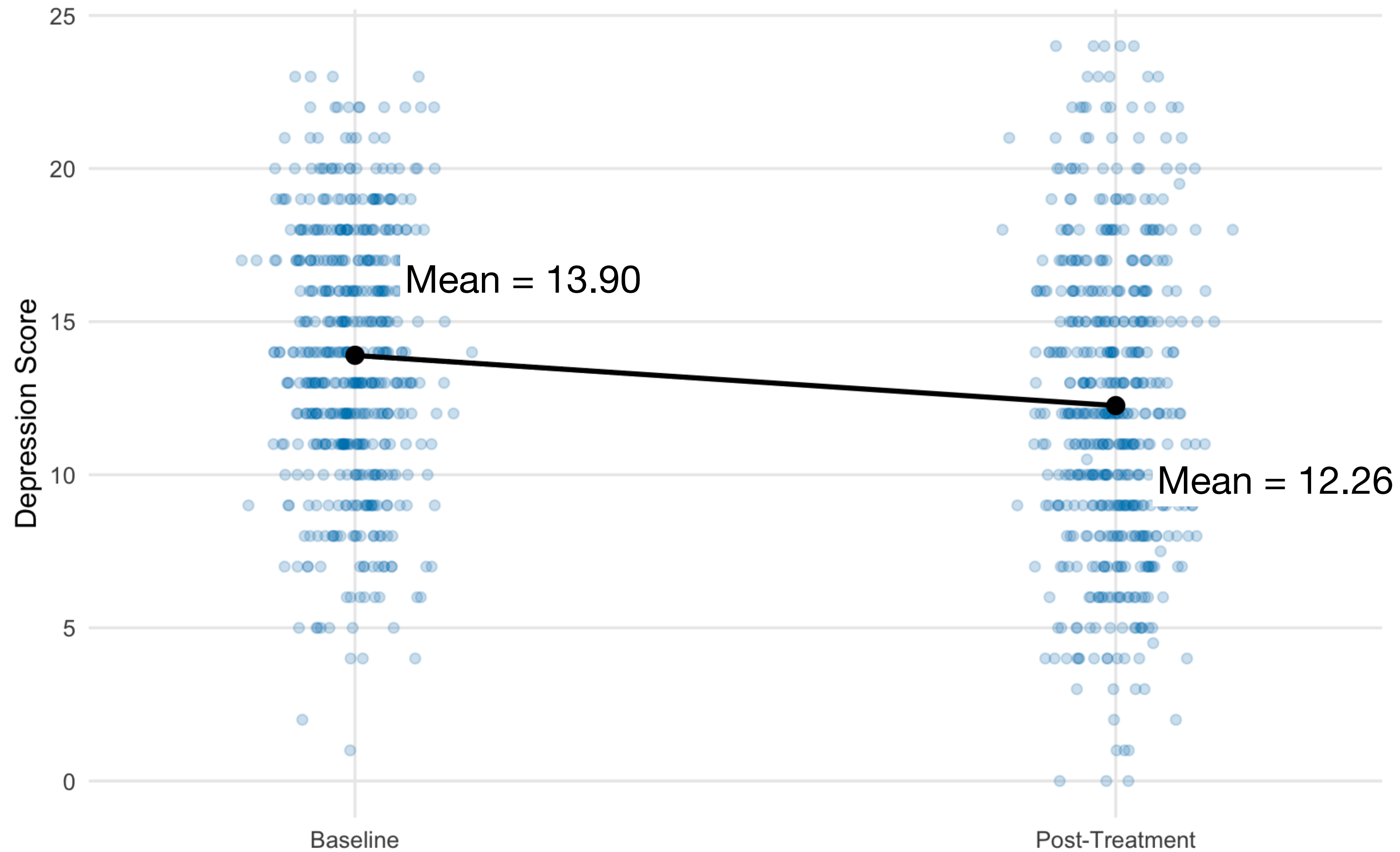
Change Is A Slope



Reminder

depression $\sim 1 + \text{timepoint}$

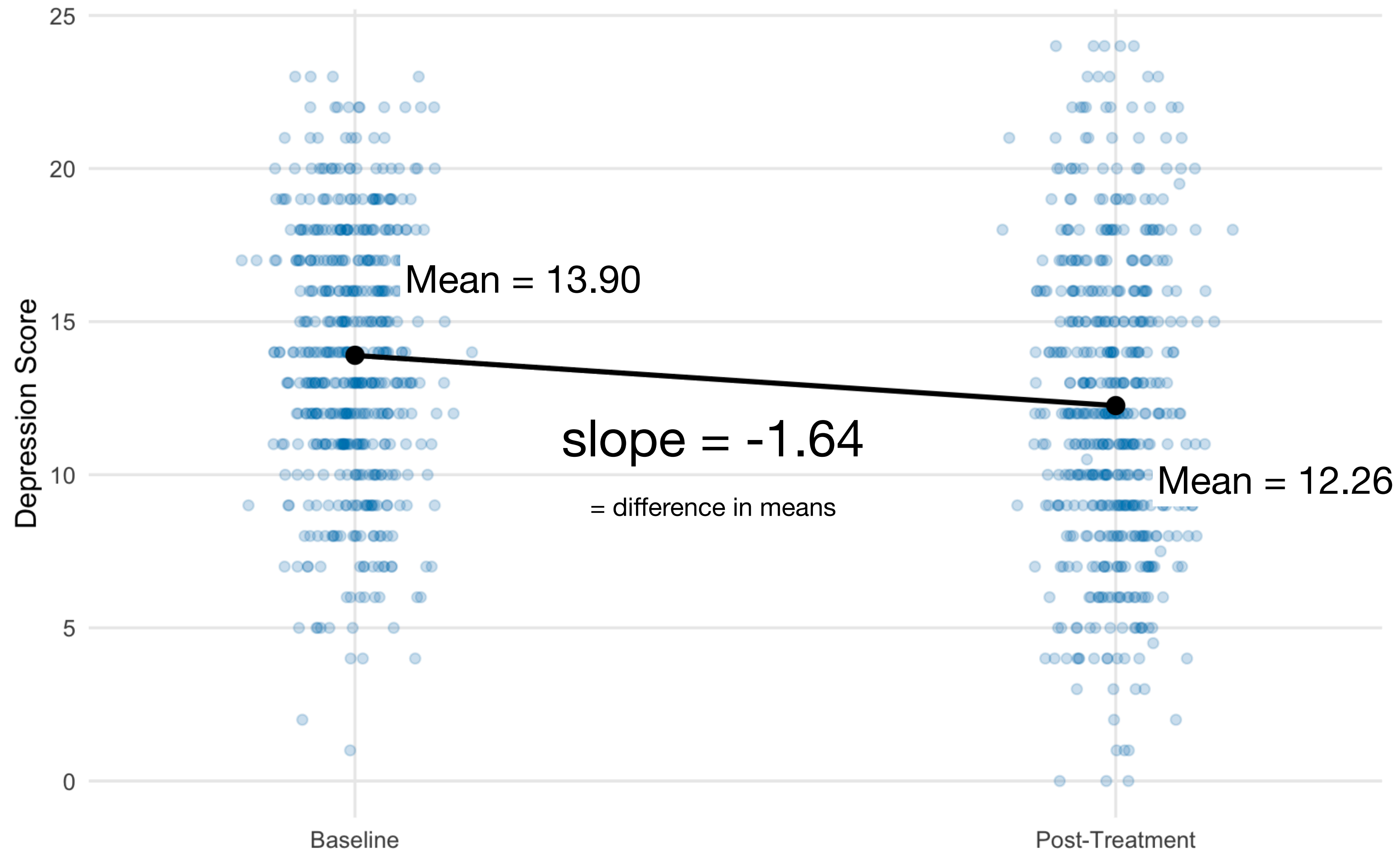
Change Is A Slope



Reminder

depression $\sim 1 + \text{timepoint}$

Change Is A Slope

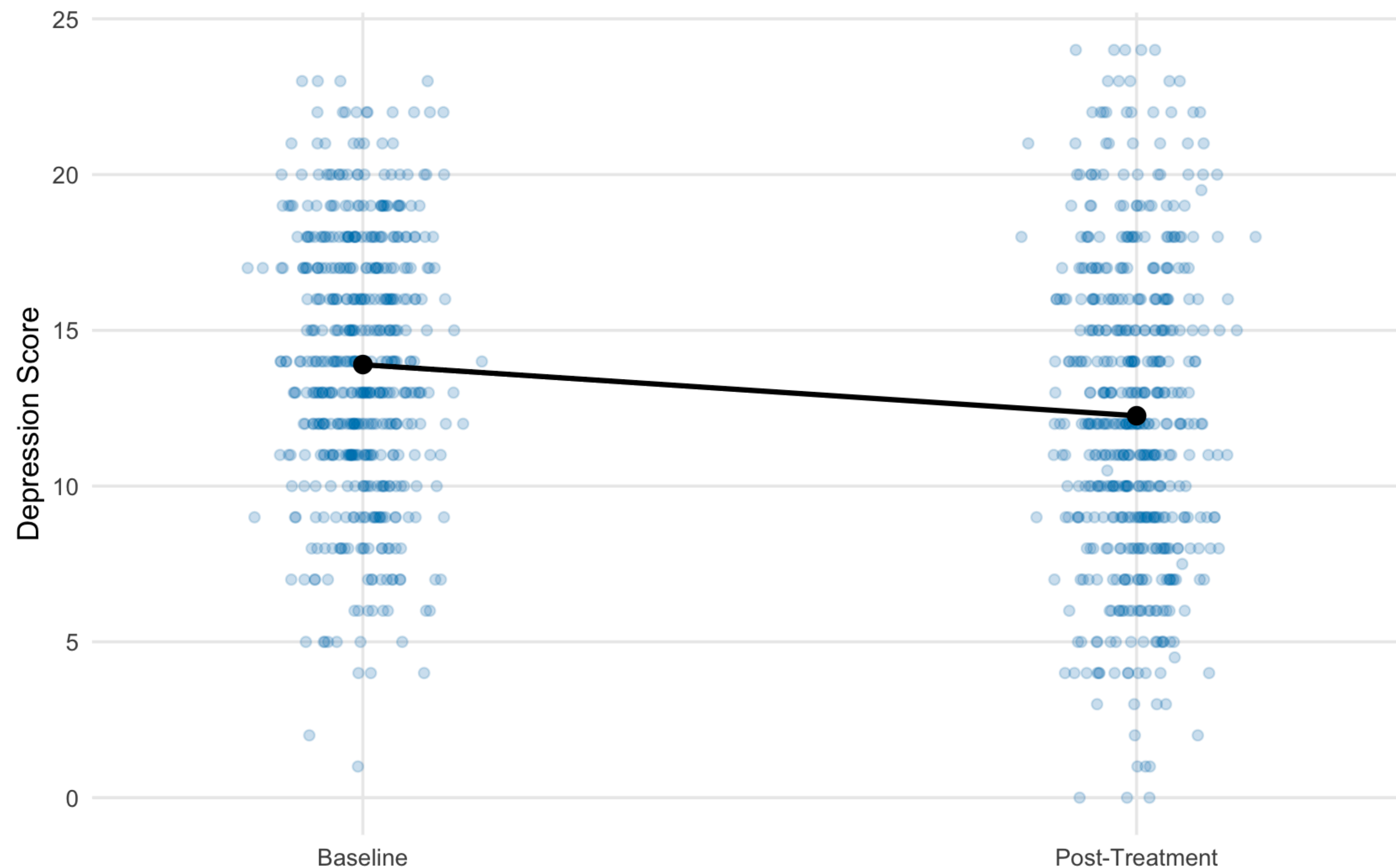


Change Is A Slope



```
data: real_data$post_treatment_depression_score and real_data$baseline_depression_score
t = -8.2685, df = 506, p-value = 1.204e-15
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -2.033390 -1.252606
sample estimates:
mean difference
 -1.642998
```


Evaluating Change



Suppose these data are from a new therapy.

Can we conclude, from these data, that our therapy is effective in reducing depression scores?

Why/why not?

Evaluating Change: Counterfactual Thinking

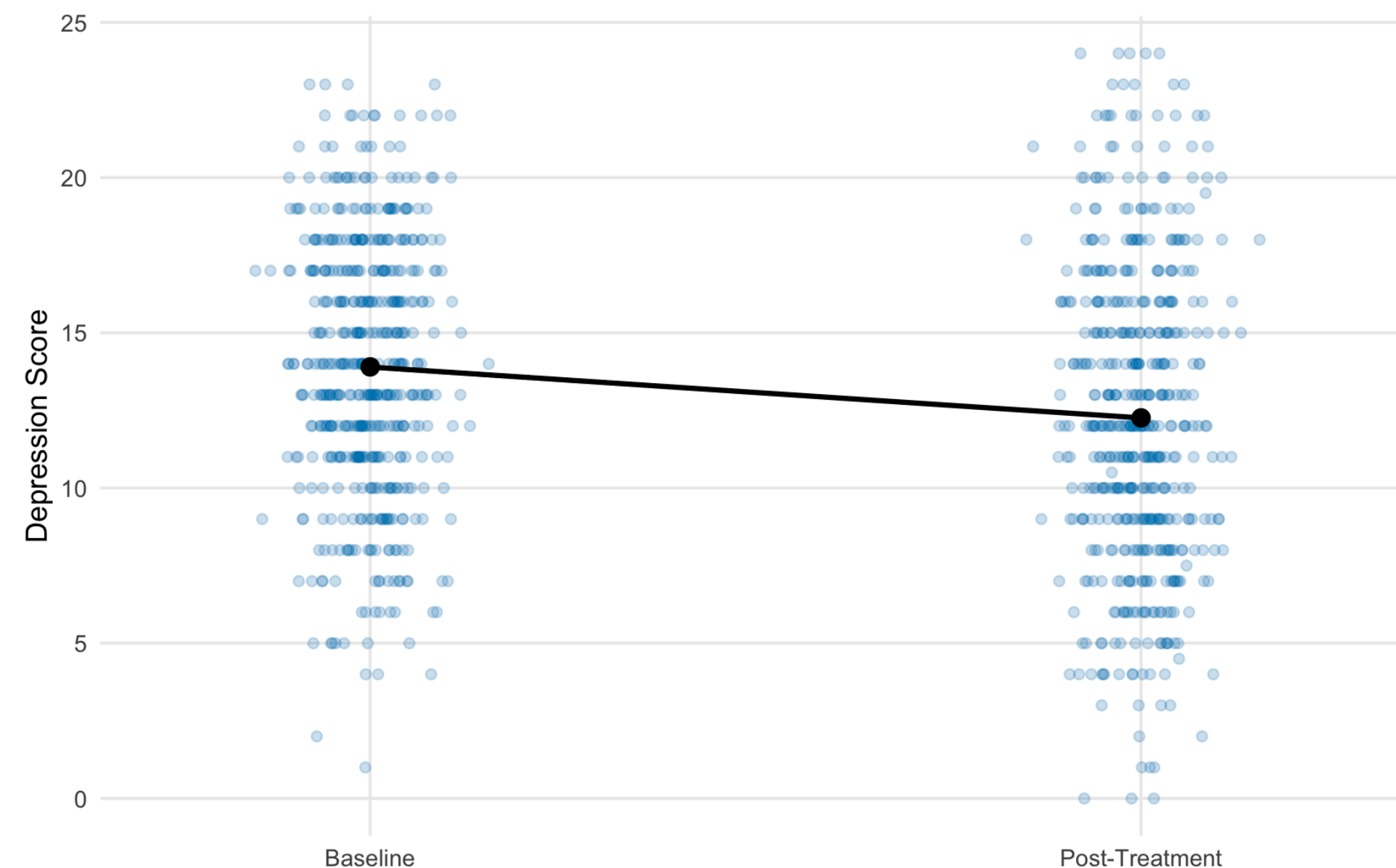
We cannot infer change based on simple **pre-post differences**.

This is because we do not know the **counterfactual**: what would have happened *without* the intervention?

Evaluating Change: Counterfactual Thinking

We cannot infer change based on simple **pre-post** differences.

This is because we do not know the **counterfactual**: what would have happened *without* the intervention?



The data we saw previously are from a real paper where no intervention was administered!

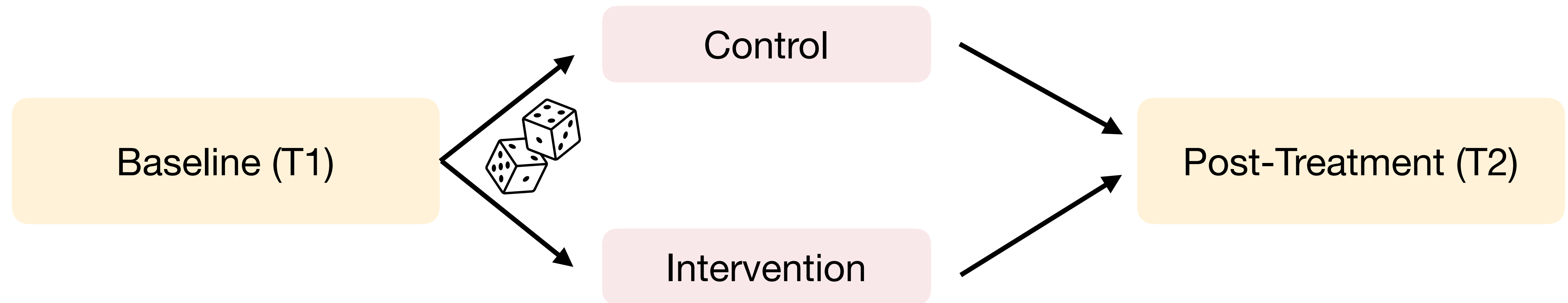
Evaluating Change: Counterfactual Thinking

To evaluate a therapy we need:

- A counterfactual comparison (AKA a control group)
- Randomization of participants to each treatment

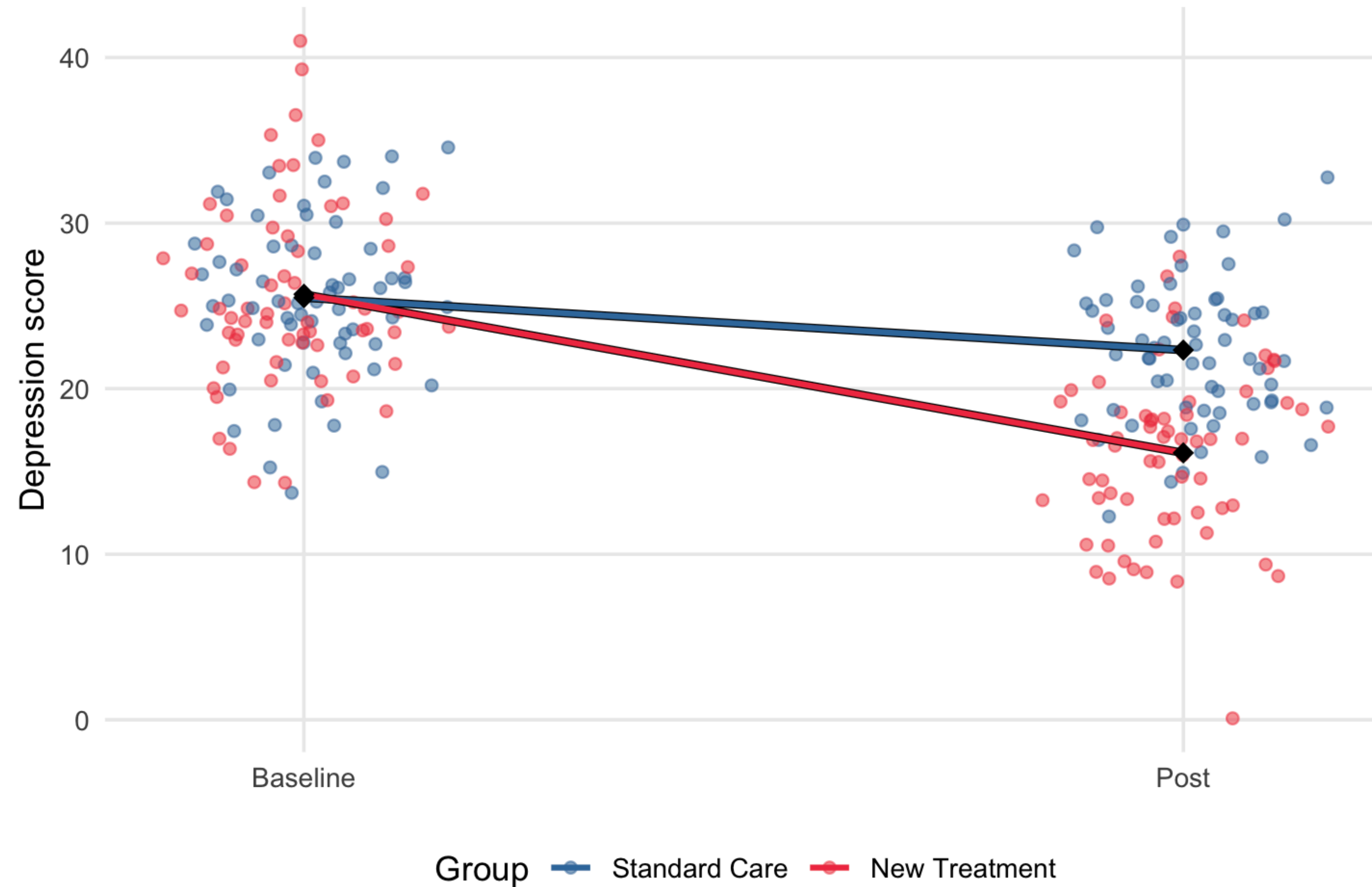
Randomized Controlled Trial

Randomized Controlled Trial



Do scores in our outcome (for example, depression scores), from baseline to post-treatment, **change more** in the intervention compared to the control?

Randomized Controlled Trial



If the slope is the change,
and we want to compare
changes,

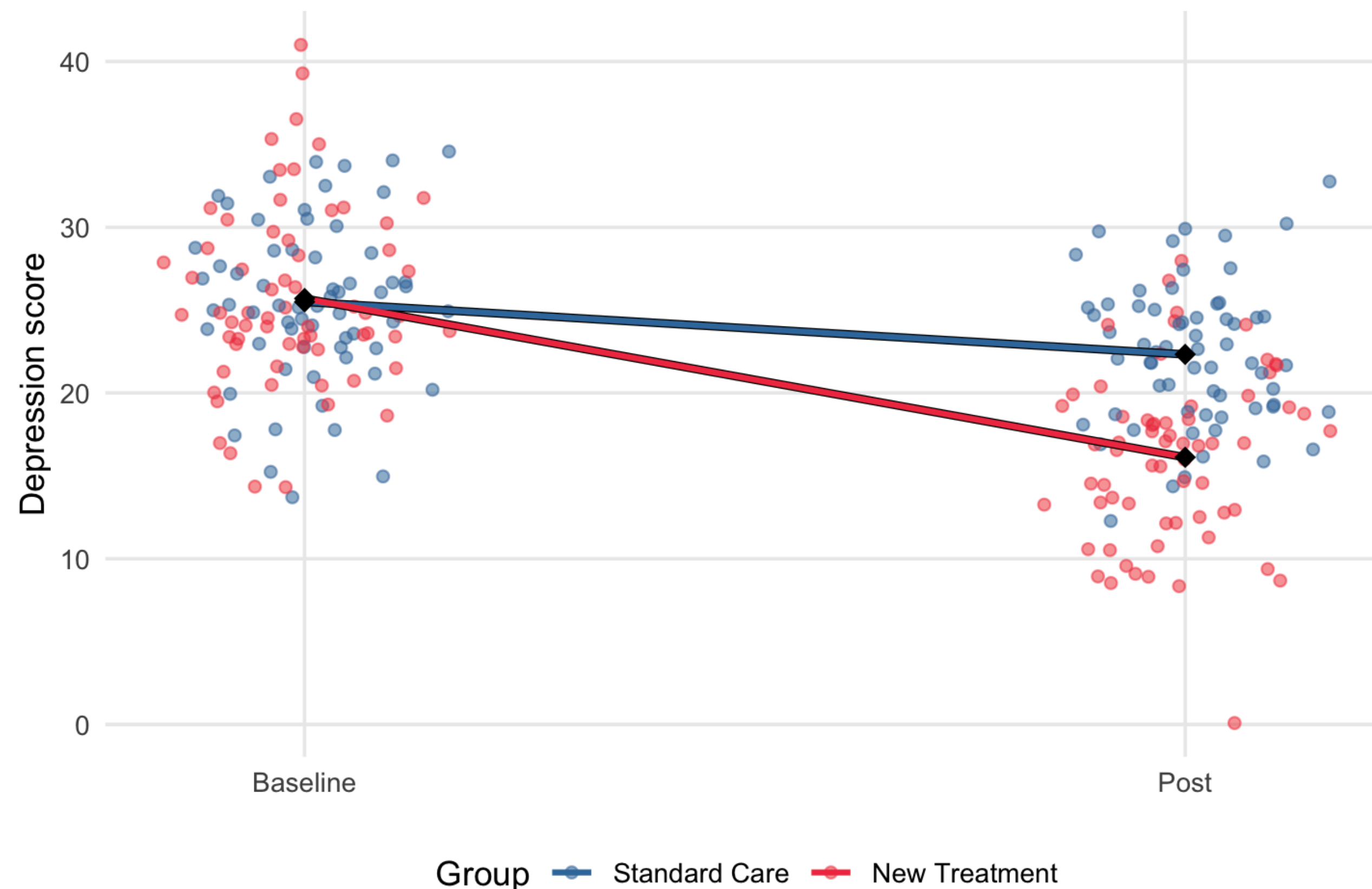
then: we can **compare
the slopes** to assess
change!

Analyzing RCT data: Mixed ANOVA

Most common approach to analysing RCT data: **mixed ANOVA**

Analyzing RCT data: Mixed ANOVA

Most common approach to analysing RCT data: **mixed ANOVA**



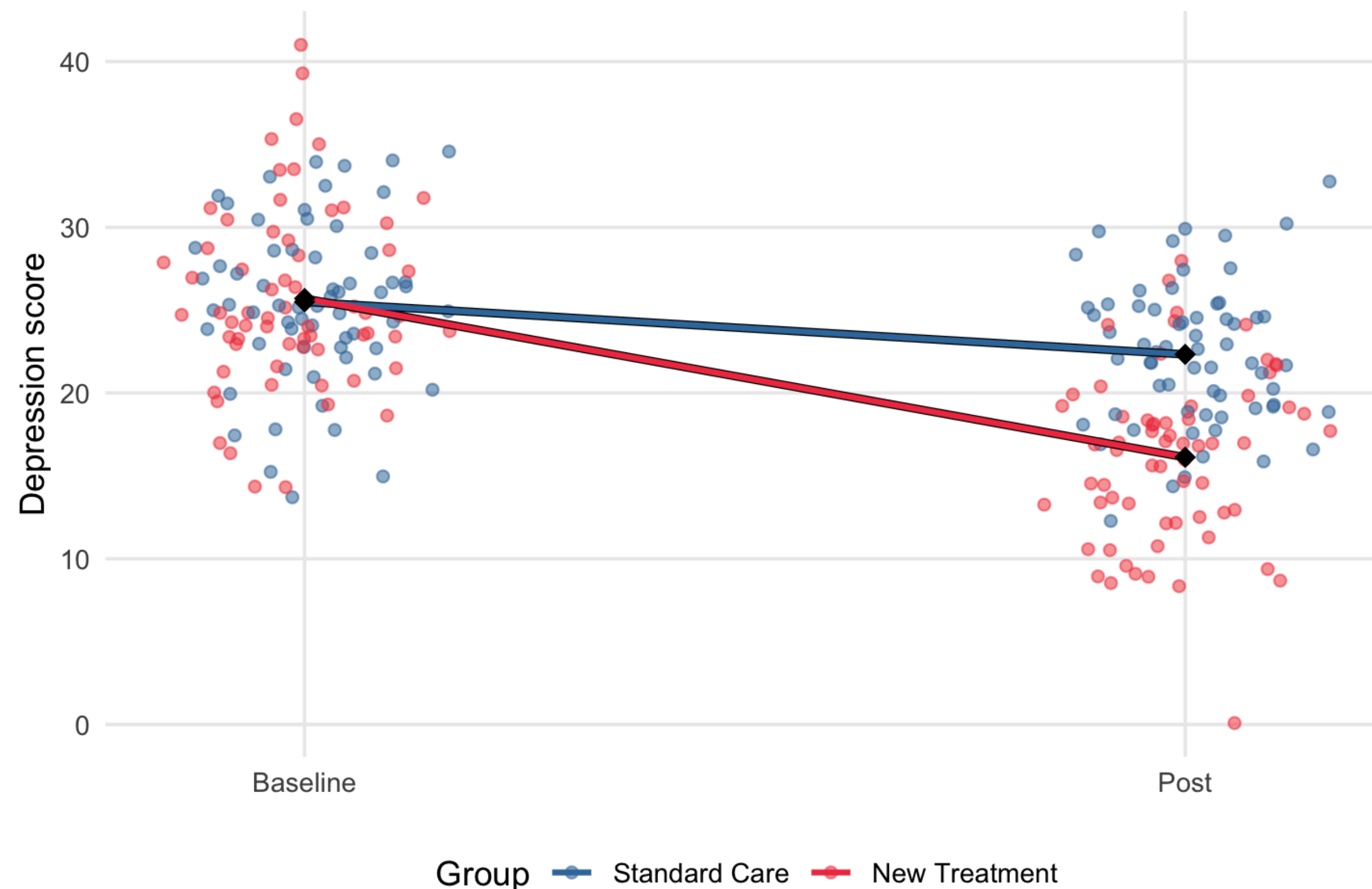
“Mixed” because there are **between** *and* **within** subject factors:

Between: intervention condition

Within: different timepoints

Analyzing RCT data: Mixed ANOVA

Most common approach to analysing RCT data: **mixed ANOVA**

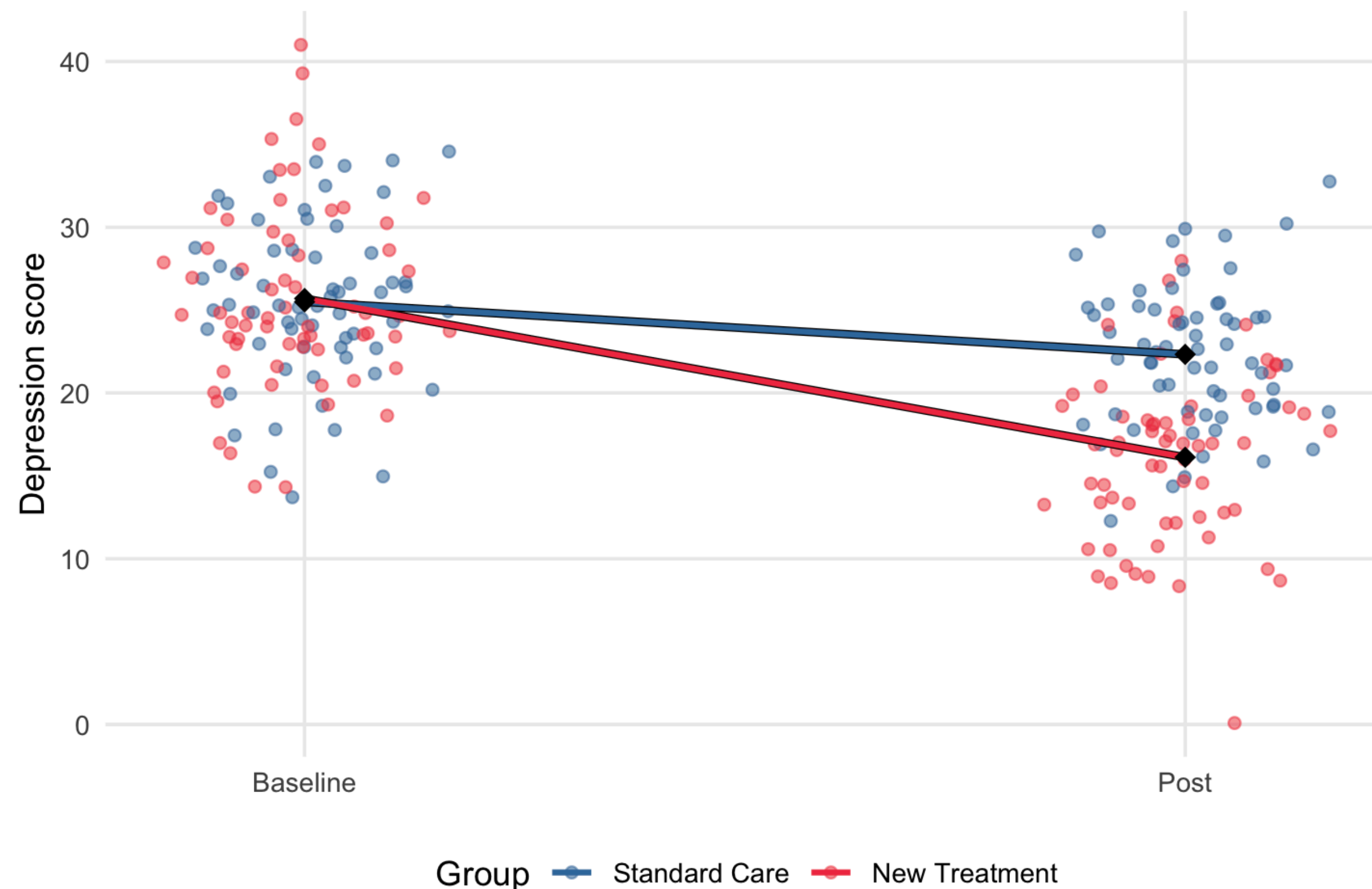


Standard mixed ANOVA formula:

$\text{depression} \sim \text{timepoint} * \text{treatment} + \text{Error}(\text{participant} / \text{timepoint})$

Analyzing RCT data: Mixed ANOVA

Most common approach to analysing RCT data: **mixed ANOVA**



Standard mixed ANOVA formula:

depression ~ timepoint * treatment +
Error(participant / timepoint)

Anova Table (Type 3 tests)

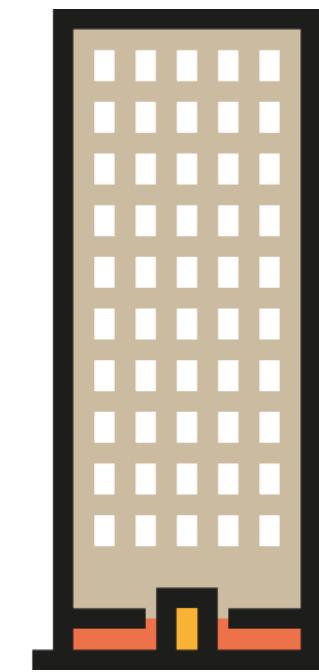
Response: y

	Effect	df	MSE	F	ges	p.value
1	group	1, 126	37.02	15.60 ***	.085	<.001
2	time	1, 126	12.29	211.63 ***	.295	<.001
3	group:time	1, 126	12.29	53.50 ***	.096	<.001

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1

What information might the mixed ANOVA be missing?

Clients might come from clusters of clinics



Clinic 1

Large clinic in city that deals with a range of different psychological conditions with a large number of clients.



Clinic 2

Small clinic in countryside that specializes in serious cases of clinical depression with small number of clients.

What information might the mixed ANOVA be missing?

Clients might come from clusters of clinics

Clients might have different therapists



Therapist 1: Ana

Private practitioner with self-referred clients. Has, on average, moderate levels of rapport with clients.



Therapist 2: Mila

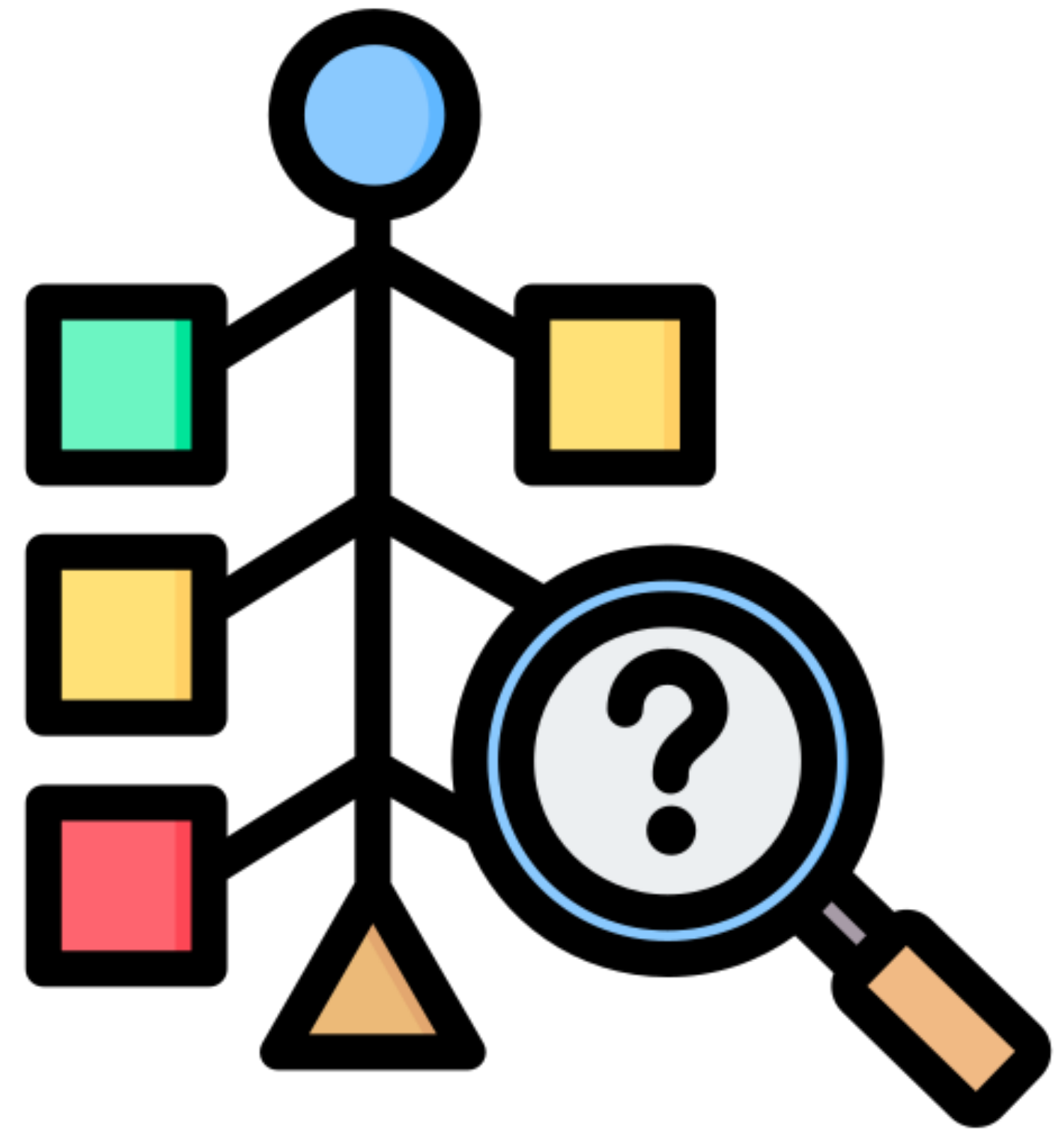
Works in a critical care clinic for severe cases of depression. Has, on average, very high levels of rapport with clients.

What information might the mixed ANOVA be missing?

Clients might come from clusters of clinics

Clients might have different therapists

Many other possible types of clustering



What information might the mixed ANOVA be missing?

Clients might come from clusters of clinics

Clients might have different therapists

Many other possible types of clustering

This violates a specific assumption of ANOVA:

Independence of observations

Our Worked Example: Different Therapists



Therapist 1: Ana

Private practitioner with self-referred clients. Has, on average, moderate levels of rapport with clients.



Therapist 2: Mila

Works in a critical care clinic for severe cases of depression. Has, on average, very high levels of rapport with clients.

Our Worked Example: Different Therapists



Therapist 1: Ana

Private practitioner with self-referred clients. Has, on average, moderate levels of rapport with clients.



Therapist 2: Mila

Works in a critical care clinic for severe cases of depression. Has, on average, very high levels of rapport with clients.

$\text{depression} \sim \text{timepoint} * \text{treatment} + \text{Error}(\text{participant} / \text{timepoint})$

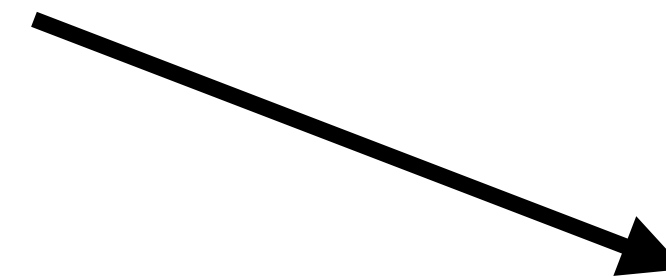
Our Worked Example: Different Therapists



Therapist 1: Ana



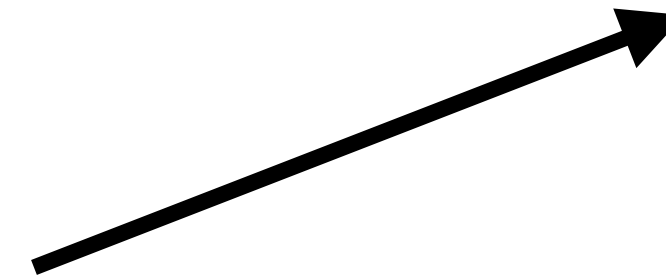
Lower mean
baseline depression
Less mean change



Therapist 2: Mila



Higher mean
baseline depression
Greater mean change

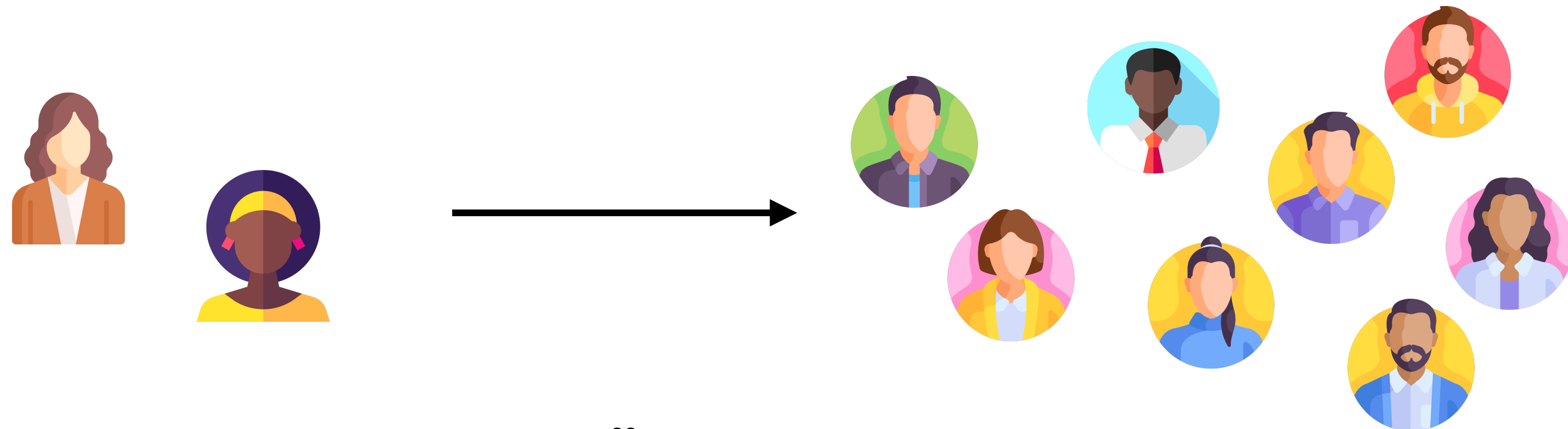


Different intercepts
Different slopes

Handling Sources of Variation

The slopes and intercepts for our therapists are different to our slopes and intercepts for treatment or timepoint

Ana and Mila are just two therapists of a **random sample** of the population of therapists who might deliver the intervention in the future



Handling Sources of Variation

Our treatment and timepoints are **fixed** effects

But our therapists are **random** effects

Handling Sources of Variation

depression ~ 1 + timepoint * treatment + (1 | therapist)

Fixed effects

Random effects

This is called a **multilevel model**

(or linear mixed model, or mixed-effects model)

Handling Sources of Variation

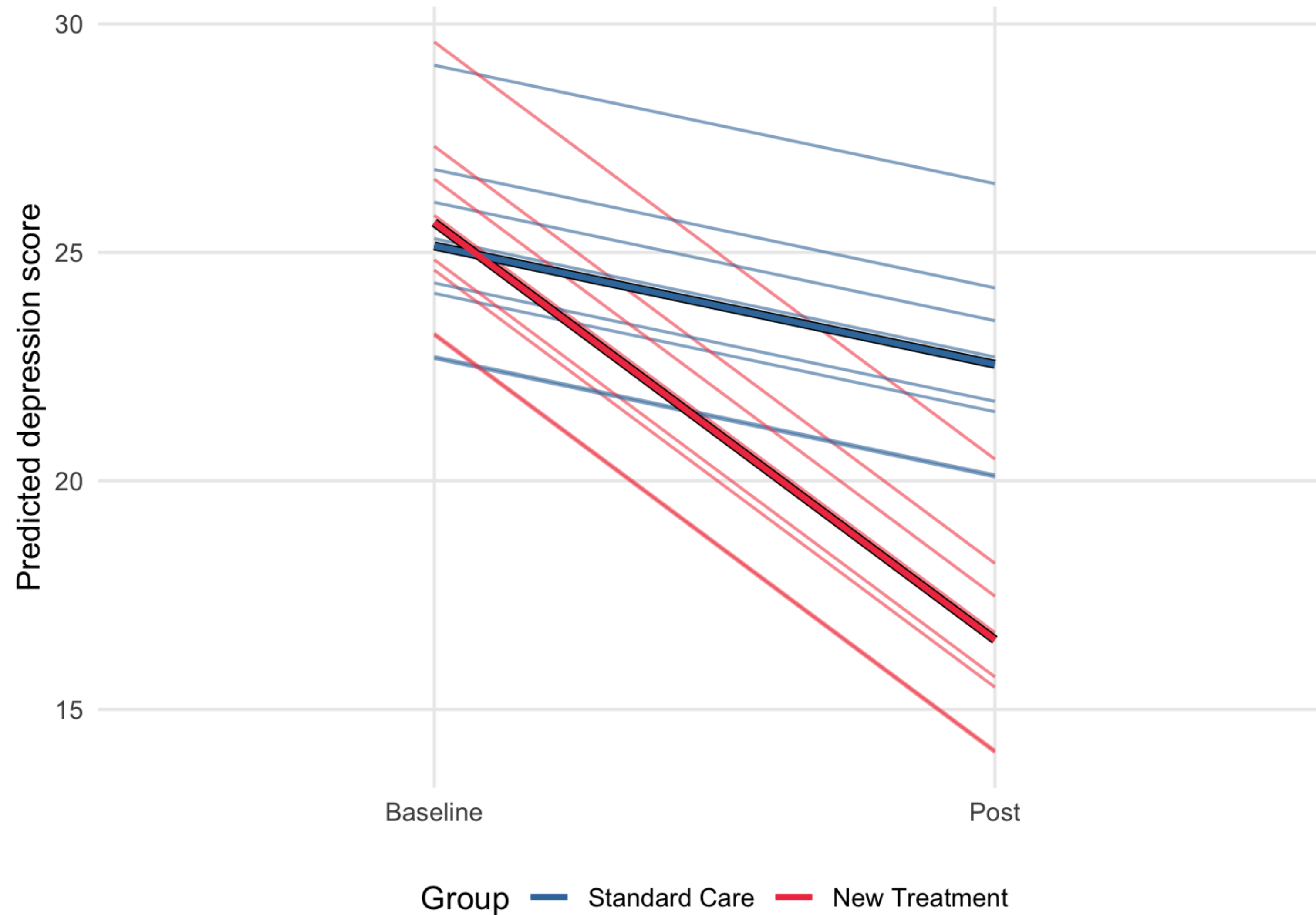
We have talked about intercepts and slopes for fixed effects.
There are also intercepts and slopes for random effects.

Random intercepts mean different levels of a random effect variable may have different means.

Random slopes model not only differences in means, but also differences in how much they change over time.

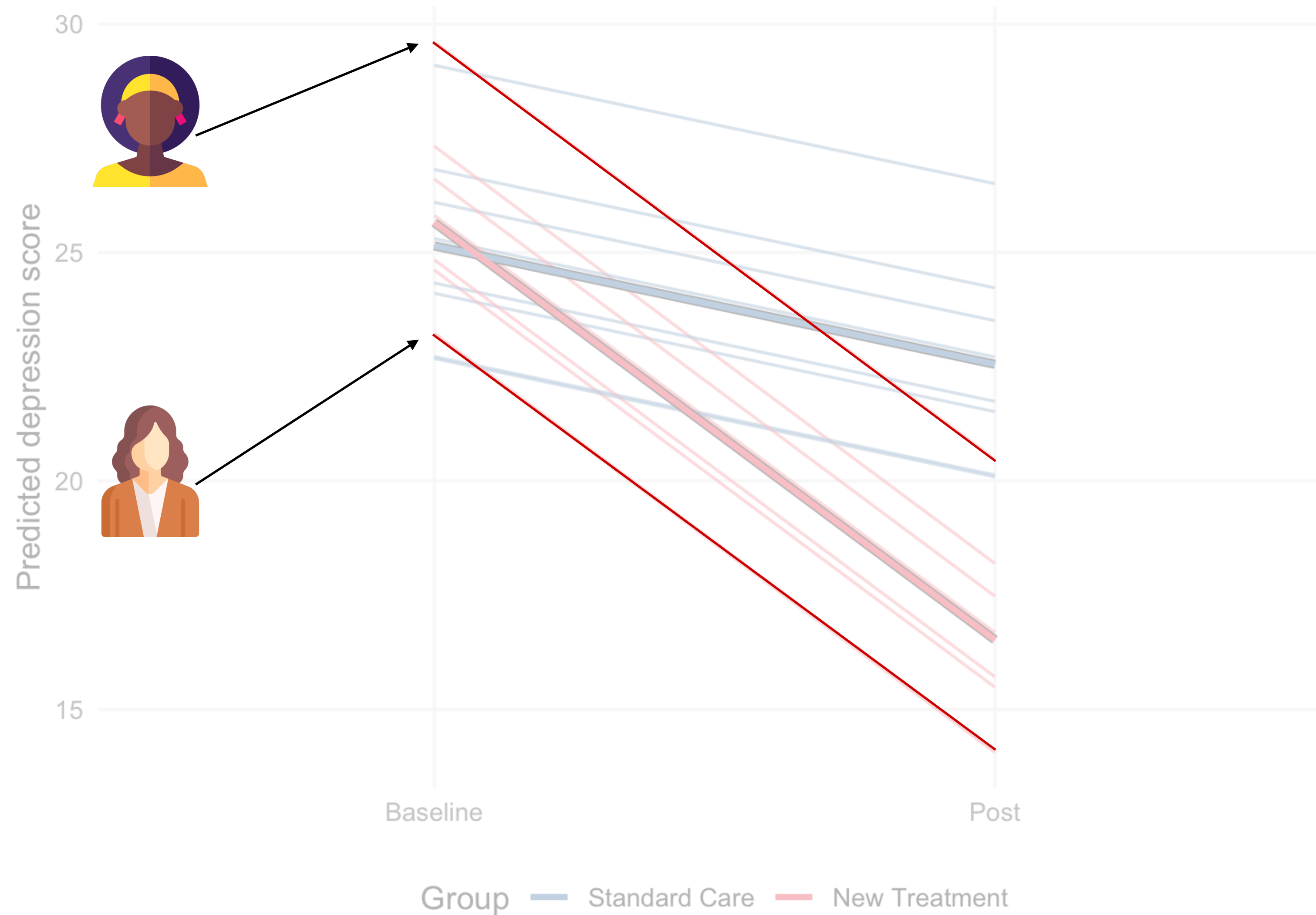
depression $\sim 1 + \text{timepoint} * \text{treatment} + (1 | \text{therapist})$

Random Intercepts



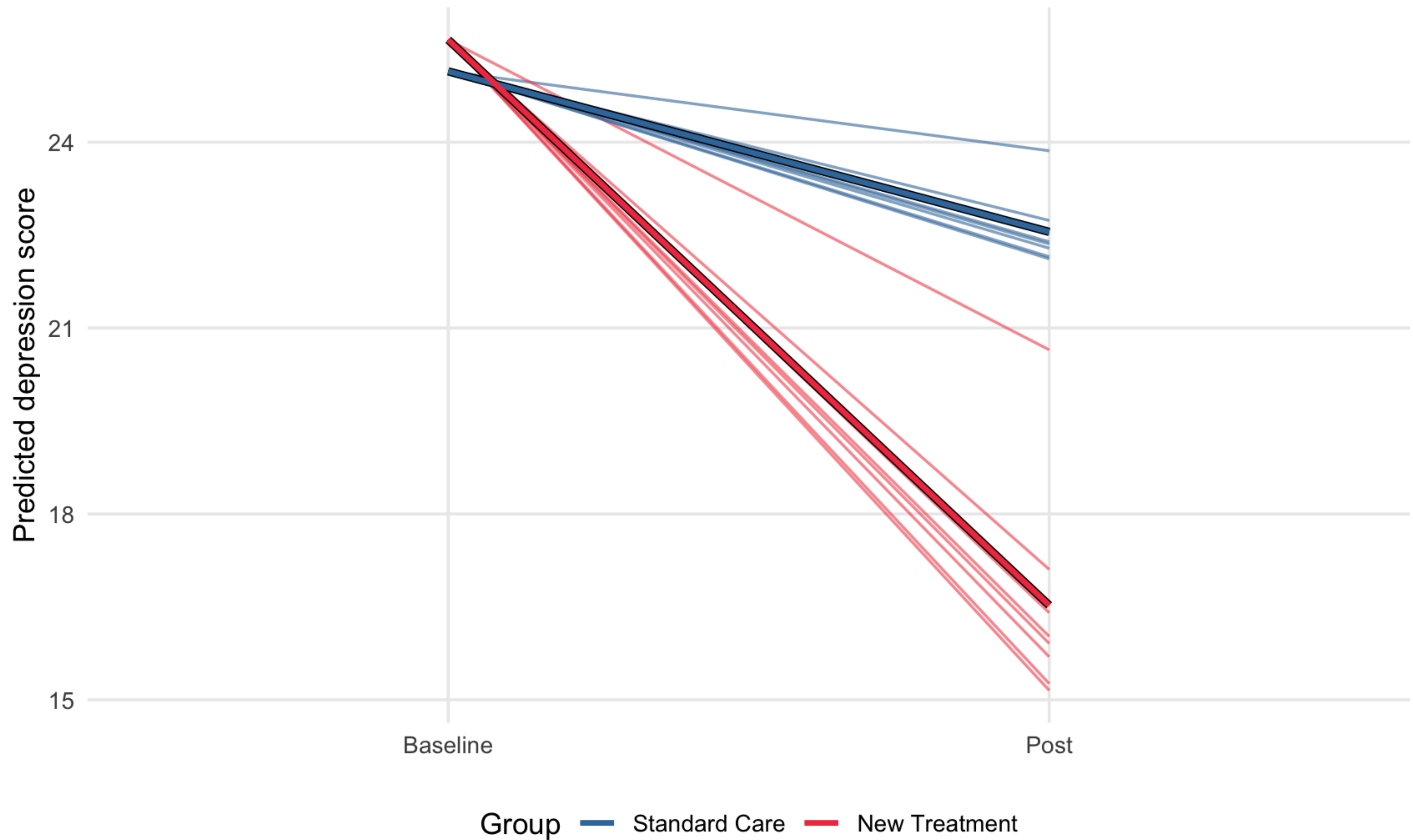
$$\text{depression} \sim 1 + \text{timepoint} * \text{treatment} + (1 | \text{therapist})$$

Random Intercepts



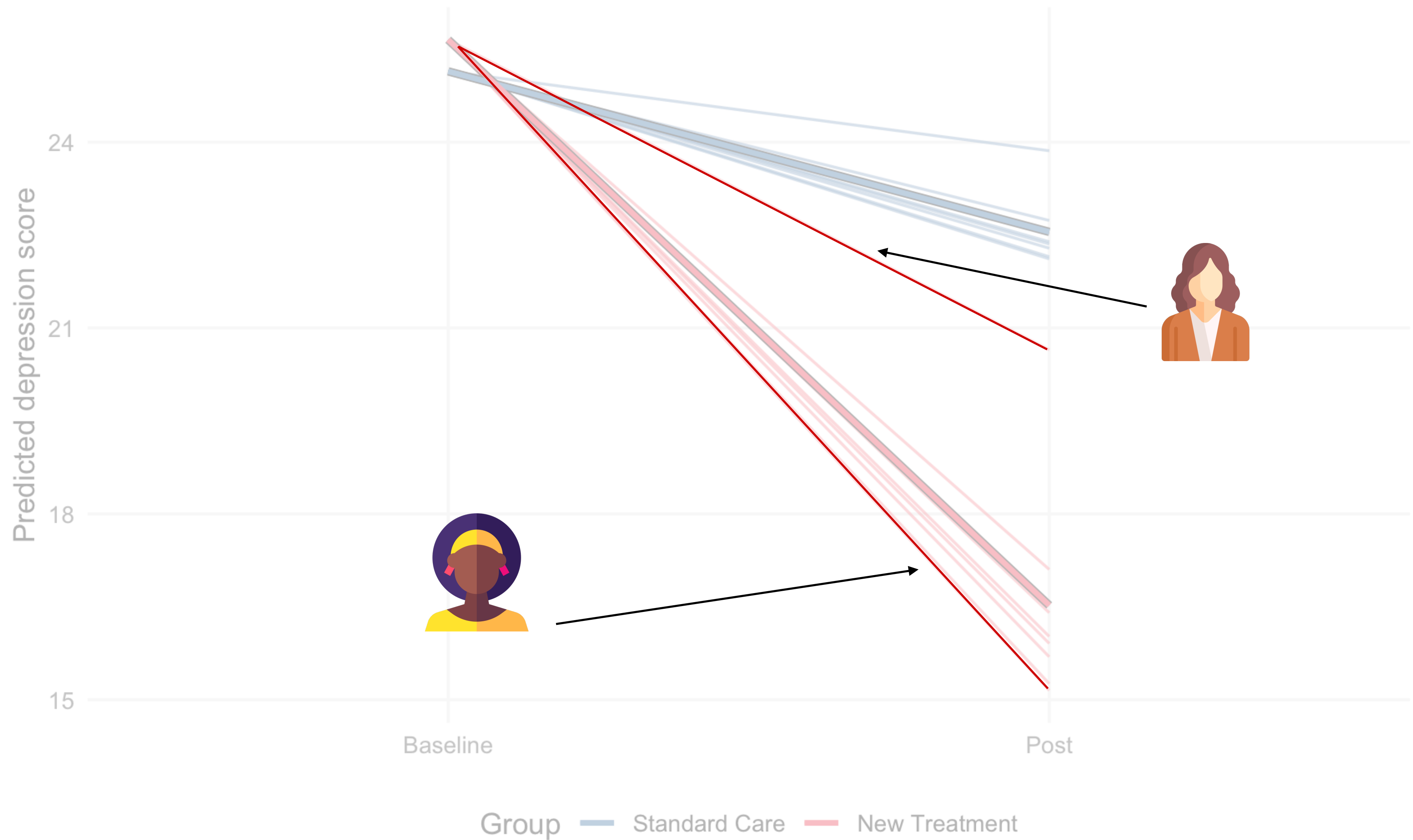
depression ~ 1 + timepoint * treatment + (0 + timepoint | therapist)

Random Slopes



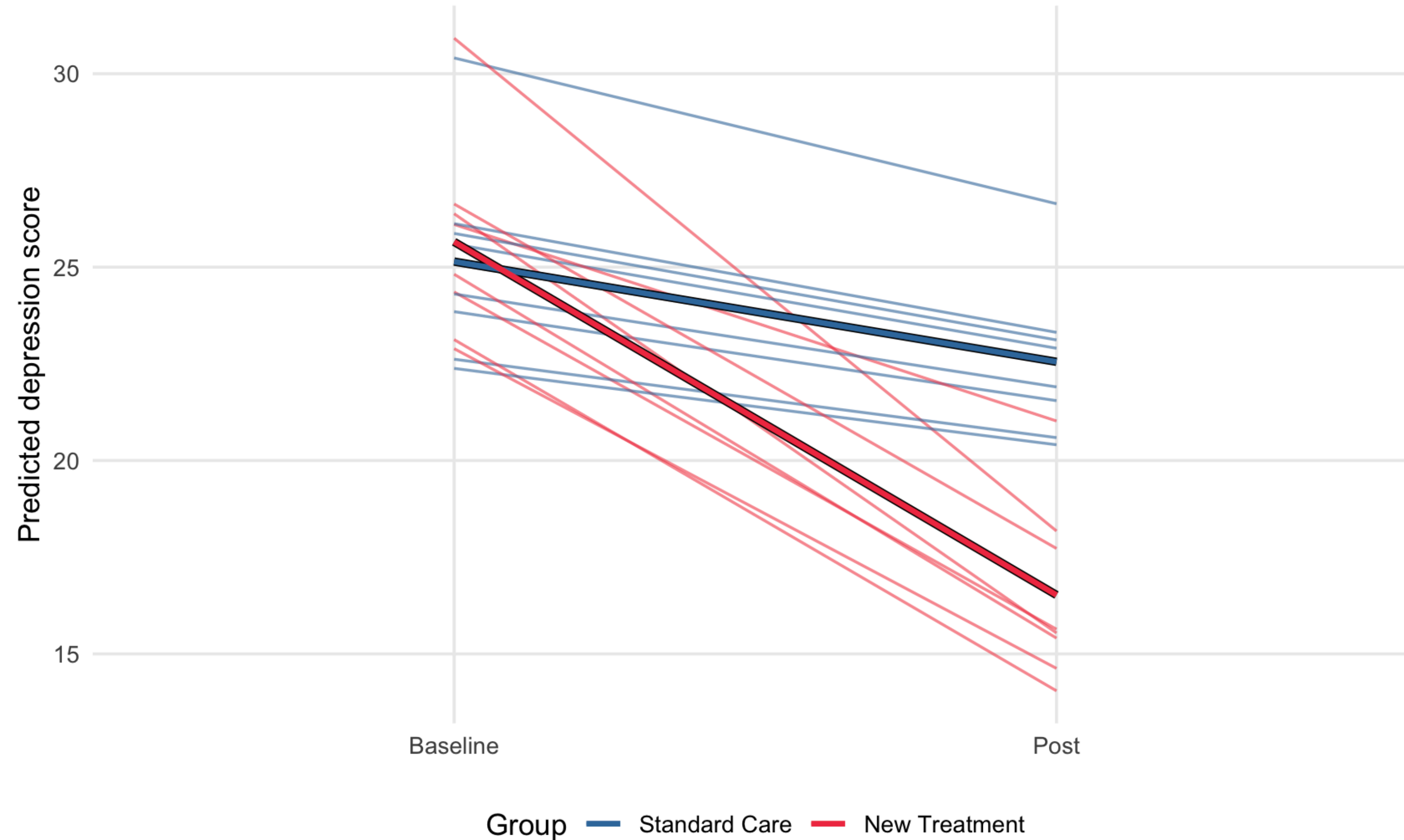
$$\text{depression} \sim 1 + \text{timepoint} * \text{treatment} + (0 + \text{timepoint} | \text{therapist})$$

Random Slopes



depression ~ 1 + timepoint * treatment + (1 + timepoint | therapist)

Random Intercepts and Slopes



$$\text{depression} \sim 1 + \text{timepoint} * \text{treatment} + (1 + \text{timepoint} | \text{therapist})$$

Random Intercepts and Slopes



Mixed ANOVA and MLMs

For the two-group, two-timepoint design,

$\text{depression} \sim \text{timepoint} * \text{treatment} + \text{Error}(\text{participant} / \text{timepoint})$

is equivalent to

$\text{depression} \sim 1 + \text{timepoint} * \text{treatment} + (1 \mid \text{participant})$

when:

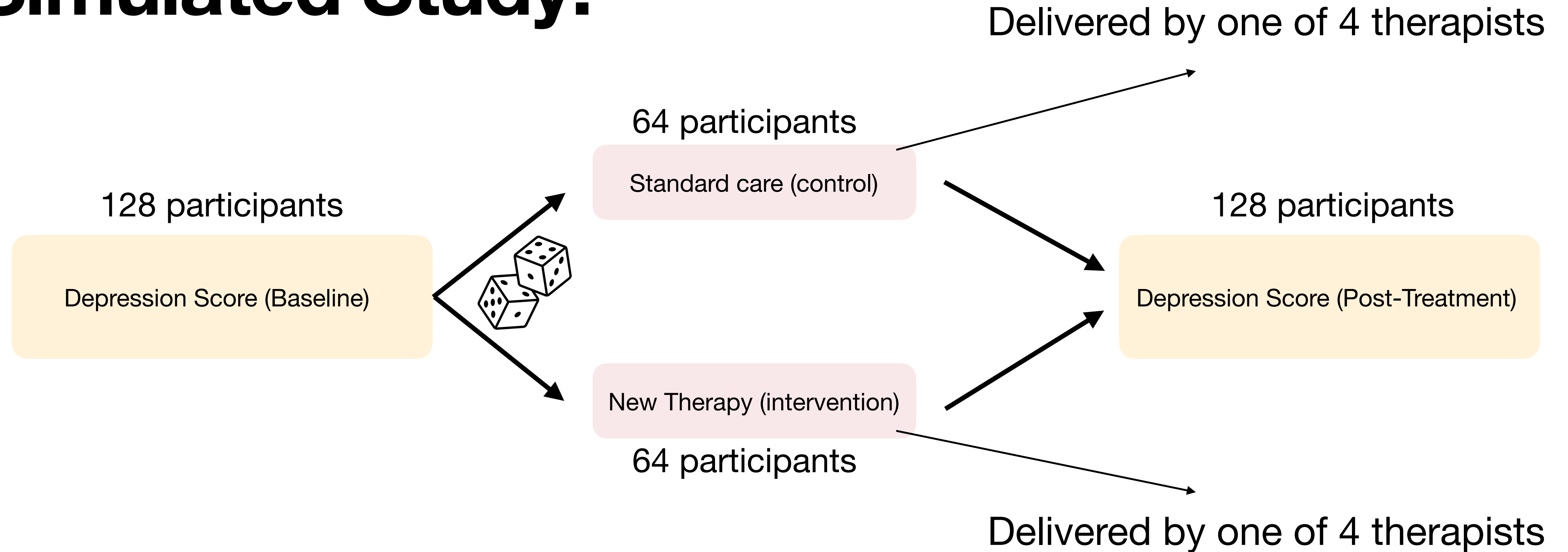
1. There is no missing data
2. There is no additional source of random variation (e.g., therapist)



What difference does it make?

Let's simulate some experiments and find out!

Simulated Study:





Our Simulation:

In our simulation, we run this study 20'000 times, with the following ground truths:

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In our simulation, we run this study 20'000 times, with the following ground truths:

Our two interventions (new therapy and standard care) have the **same effect on depression** (5 point reduction).

There is a **true random intercept effect** for participants.

There is a **true random slope effect** for therapist.

Our Simulation:

We know in our simulation that the true interaction effect is zero.

How often should we expect to see $p < .05$ for the interaction effect?

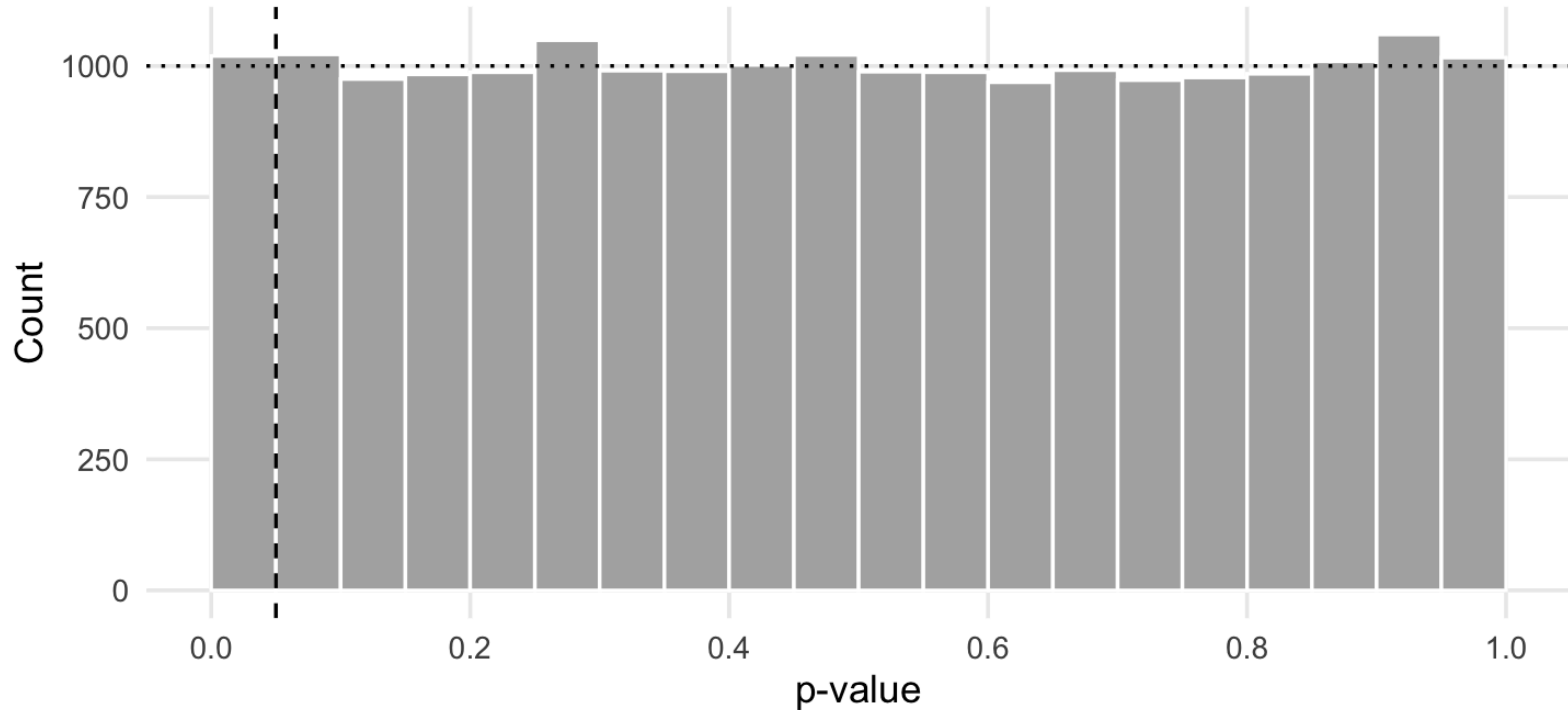
Our Simulation:

We know in our simulation that the true interaction effect is zero.

How often should we expect to see $p < .05$ for the interaction effect?

5% of cases (i.e., alpha)

p-value distribution under null



Our Simulation:

We analyse our 20'000 simulated trials with 3 models:

1. Mixed ANOVA:

$\text{depression} \sim \text{timepoint} * \text{treatment} + \text{Error}(\text{participant} / \text{timepoint})$

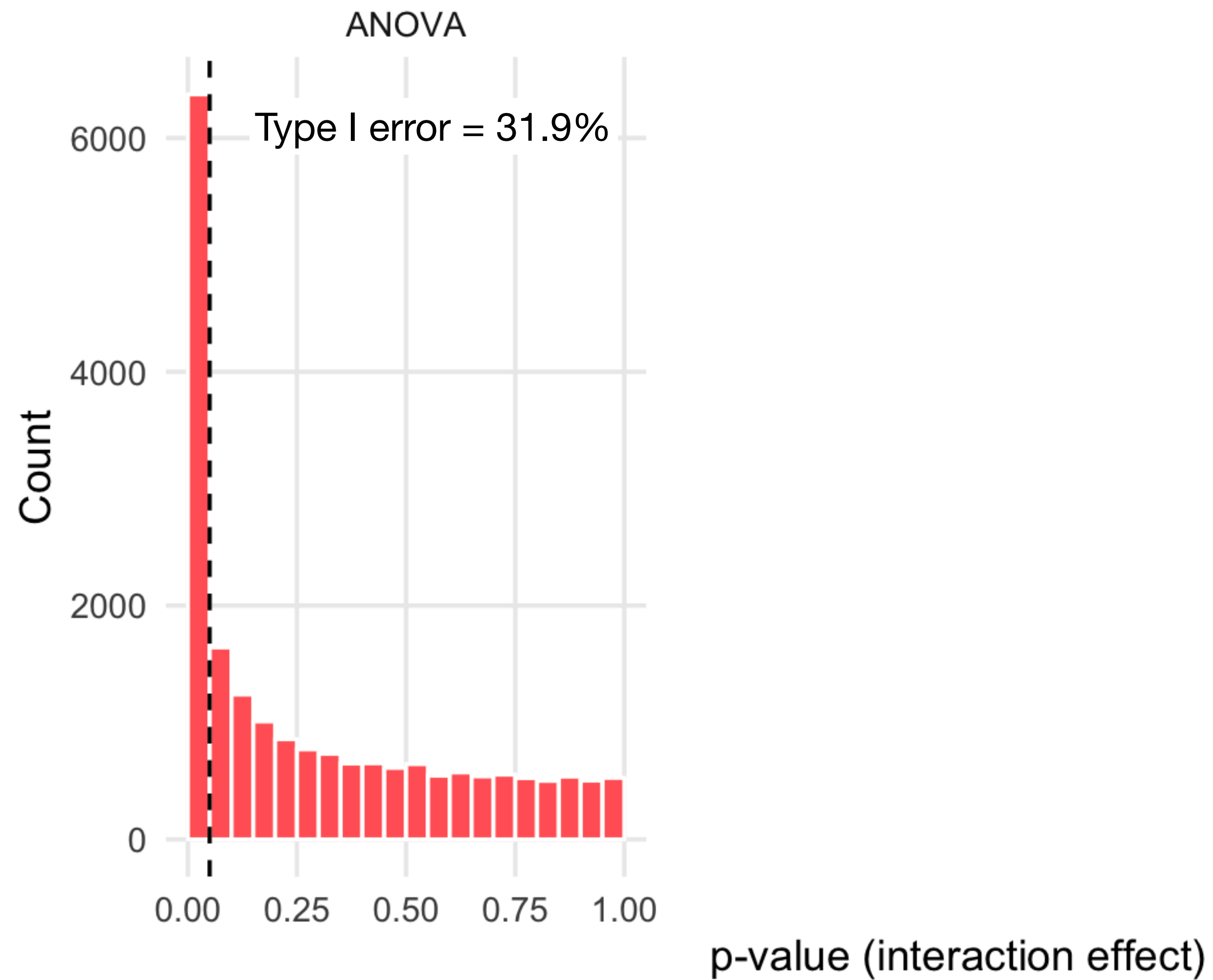
2. Multilevel model with random therapist intercept:

$\text{depression} \sim \text{timepoint} * \text{treatment} + (1 \mid \text{participant}) + (1 \mid \text{therapist})$

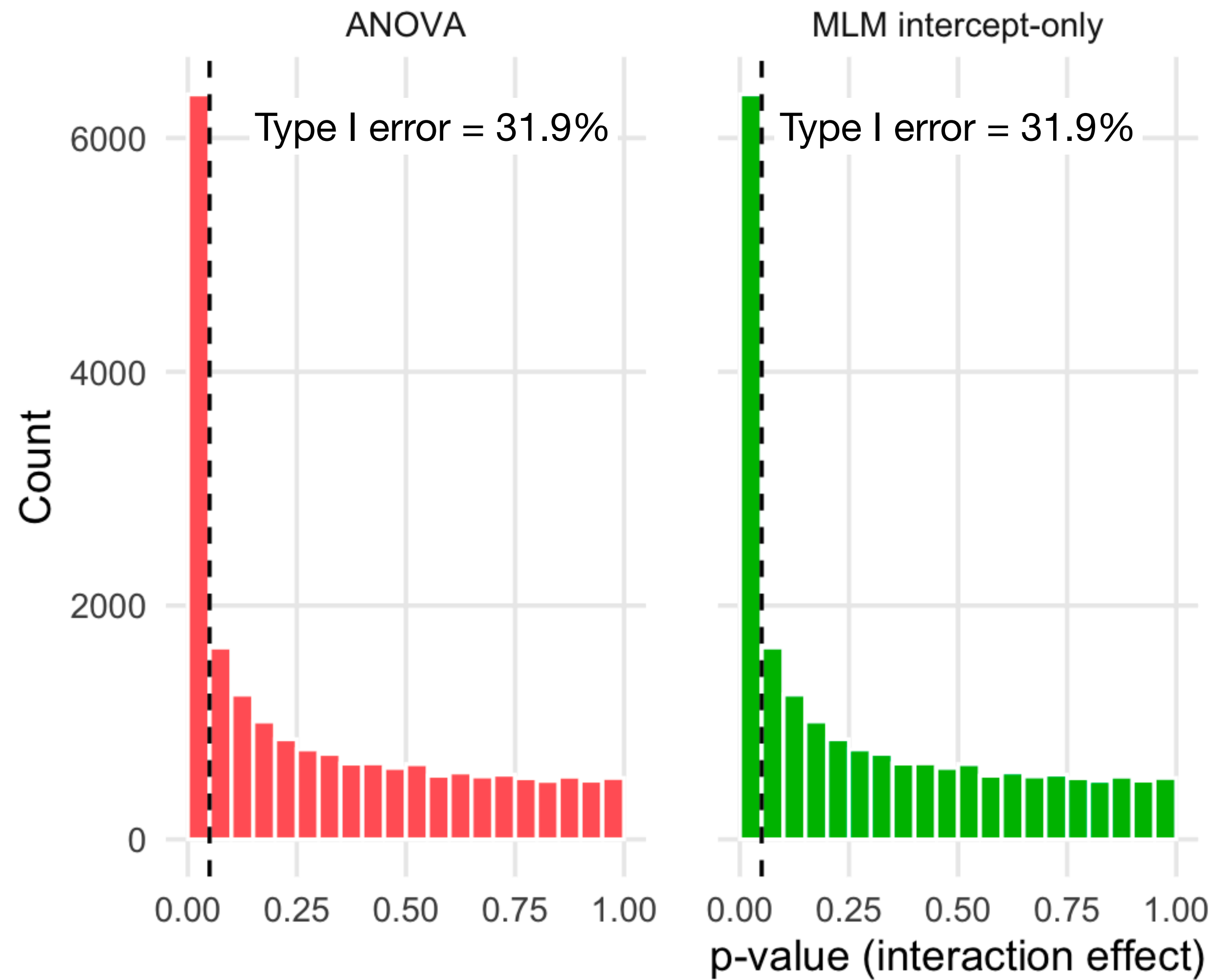
3. Multilevel model with random therapist intercept and slope:

$\text{depression} \sim \text{timepoint} * \text{treatment} + (1 \mid \text{participant}) + (1 + \text{timepoint} \mid \text{therapist})$

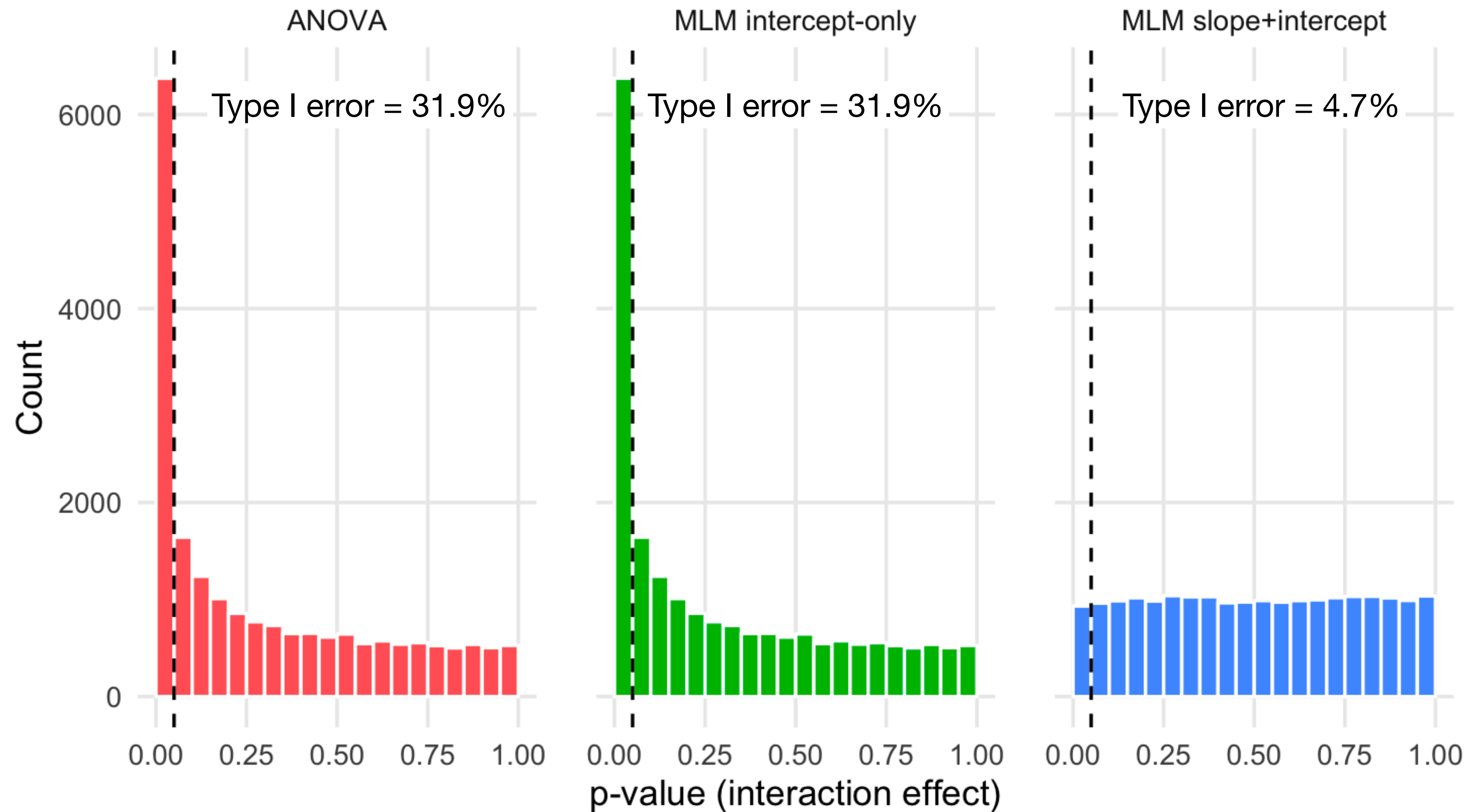
Our Simulation:



Our Simulation:



Our Simulation:



So when can we use ANOVA?

When the design is balanced
and genuinely simple

When our data is too sparse to
meaningfully measure random effects

But ANCOVA is better (next week)

Summary

Mixed ANOVA **ignores clustering** which can frequently occur in RCT data collection

Multilevel models are **much more flexible** and can therefore much more easily represent **a more accurate model** of the world

Models that do not account for important clustering, including mixed ANOVA, can lead to **increased Type I error rates**

Practice Exam Questions



PREPARE

PREPARE

MY COURSES

Intro stats42

Adv methods28

Causal inference19

Meta-research15

THIS WEEK

Lecture 06 — MLMs

Lecture 07 — ANCOVA & baseline

Seminar — RCT analysis

PREPARE — practice session

Adv methods · Lecture 06 · Multilevel models for repeated measuresQ6 / 107 day streak

Multiple choiceAnalyseHard

In a two-arm, two-timepoint therapy RCT, patients are nested within therapists ($ICC \approx 0.10$ due to therapist effects). You simulate 10,000 datasets *under the null* of no treatment effect and analyse each one with a 2×2 mixed ANOVA that ignores therapist clustering. What does the resulting distribution of p-values for the group $\times$ time interaction look like?

FROM THE LECTURE

"Under a true null with no clustering, p-values are uniform on $[0, 1]$ and the test is nominal at α . Once a clustering source like therapist is ignored, the residual variance is mis-partitioned and the sampling distribution of the test statistic no longer matches its reference..."

A

Approximately uniform on $[0, 1]$ — clustering inflates standard errors symmetrically, so the false-positive rate stays at the nominal 5%.

B

Right-skewed with an excess of small p-values — ignoring therapist clustering treats correlated observations as independent, so Type I error is inflated above 5%.

C

Left-skewed with an excess of large p-values — clustering reduces effective sample size, making the test conservative and Type I error fall below 5%.

D

Uniform overall but with a spike near $p = 1$ — the within-subjects error term in mixed ANOVA already absorbs therapist variance, so the test remains calibrated.

Generated from

Lecture 06 transcript · 31:40–34:12

slide 22

Flag question

Skip

Submit answer

Synced 2 min ago

SESSION

Correct

Avg time

Confidence

Est. mastery

CONCEPTS TESTED

null distribution

clustering

ICC

Therapist

mixed ANOVA

54

Revision

Mixed ANOVA vs Linear Mixed Models — interactive simulator

Simulates many 2×2 therapy trials with therapists nested in arms (each therapist delivers only one protocol). Fits all three analyses to each trial and plots the resulting p -values for the time $\times$ group interaction. Set the true treatment effect to 0 to test calibration (false-positive rate); set it non-zero to test power.

Number of simulations

100

500

2,000

Therapists per arm

2

4

15

Patients per therapist

4

16

30

True treatment effect (interaction)

-10

0

10

0 = no real effect → rejection rate IS the false-positive rate.

Therapist variability — pre-post change (SD)

0

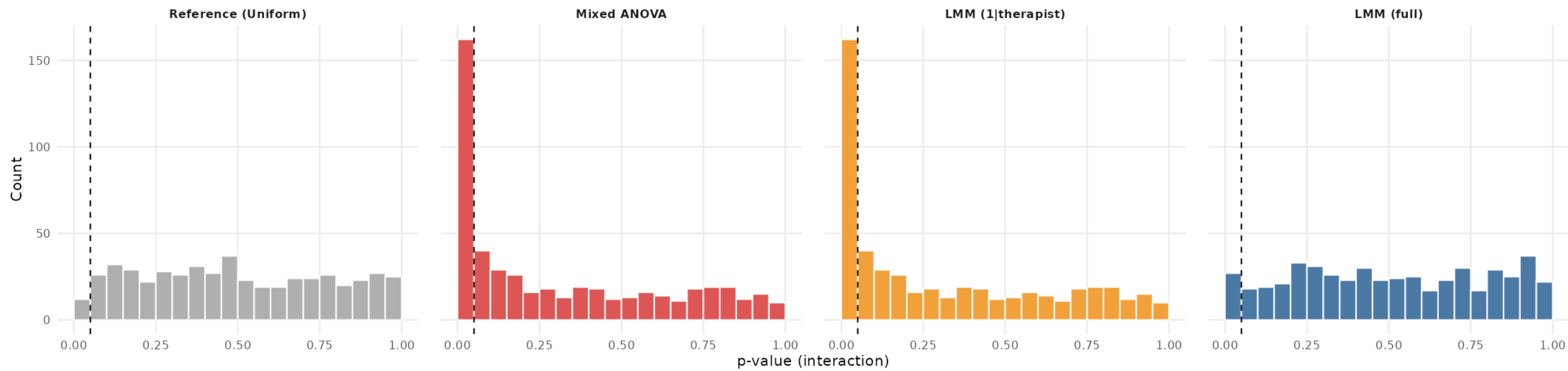
2

5

Run simulation

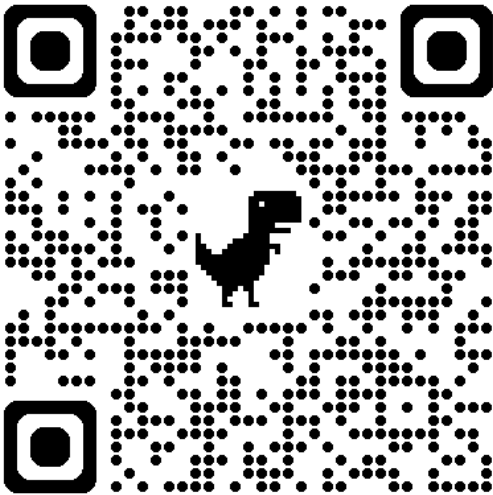
Each simulated trial: 128 patients (64 per arm, 4 therapists per arm).

Method	Rejection rate ($p < .05$)	Trials used
Mixed ANOVA	0.324	500
LMM (1 therapist)	0.324	500
LMM (full)	0.054	500



What you're seeing: the true treatment effect is set to 0.00, so each rejection rate above is a measure of *false-positive rate*. A correctly-calibrated test should sit at .05.

- **Mixed ANOVA:** 0.324
- **LMM (1|therapist):** 0.324
- **LMM (full):** 0.054



Reading

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simulations.tidyverse.se

In R and tidyverse



Understanding Statistics

through Monte Carlo simulations

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Next Week

Examine impact of **mixed ANOVA vs. ANCOVA** for modeling RCT data

How to know **which random effects** to include in our models

Introduce the idea of **good and bad controls** from a **causal inference** perspective

Thank You!

Examine impact of **mixed ANOVA vs. ANCOVA** for modeling RCT data

How to know **which random effects** to include in our models

Introduce the idea of **good and bad controls** from a **causal inference** perspective